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RESEARCH PAPER

Effectiveness of product return-prevention instruments: Empirical evidence

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Abstract The convenience and ease of online shopping reduce consumers' risk perceptions, which encourages the continued growth of online retailing but also may force online retailers to deal with expensive and excessively high product return rates. Despite efforts by e-commerce management practitioners and scholars to identify determinants of customer product return behavior, scarce research investigates the effectiveness of instruments designed explicitly to reduce customers' actual return rates. Drawing on risk theory, this article tests the influence of three important instruments on product return prevention. Three separate field experiments among customers of a well-known European online retailer reveal, unexpectedly, that the use of a money-back guarantee increases product returns, whereas product reviews decrease the product return rate. The provision of free return labels has no influence on customer product return behavior. This article concludes with some managerial and theoretical implications of these results.

Keywords E-commerce · Money-back guarantee · Online shopping · Prevention instruments · Product returns · Product reviews

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Introduction

Electronic retailing continues to grow at a faster rate than conventional retailing (MacKenzie et al. 2013; RetailMeNot 2015), especially in apparel product categories (Ben-Shabat et al. 2016). For the U.S. apparel sector, online business captured $60 billion in 2015 (eMarketer 2015), and the growth trends are upward. Yet online retailers of apparel and other popular online products (e.g., consumer electronics) face a major concern: product returns (Rao et al. 2014; Walsh et al. 2016; Zhou and Hinz 2016). In the U.S., the average return rate is 30% for clothing online (Shah 2014). Moreover, it was estimated that in the U.S. in the pre-Christmas months of November and December of 2015, $20 billion worth of products bought online were returned (Maple 2015). Consumers may return non-defective products that do not fit their preferences (Shulman et al. 2010), especially those they order online. In contrast with offline shopping, online customers cannot evaluate the product before purchase by touching, smelling, or trying it on (Ofek et al. 2011; Shulman et al. 2010). For an online retailer, returns lead to costs and perhaps unsatisfied customers, even though most returned products are non-defective (Hess et al. 1996; Walsh et al. 2016). The costs associated with product returns are estimated to be around $30 billion (£20bn) for British online retailers (Ram 2016).

In response, some online retailers implement return policies aimed at reducing product returns while increasing customer value (Janakiraman and Ordóñez 2012; Petersen and Kumar 2009) or simply accept high return rates as part of their consumer-friendly, lenient return policies. For example, Zappos offers customers a 365-day, free return policy; the European apparel retailer Zalando has a 100-day, free return

policy. Although some studies suggest that lenient return policies offset the costs associated with high return rates (e.g., Janakiraman et al. 2016; Petersen and Kumar 2009; Zhou and Hinz 2016), others indicate that they lead to even more excessive product returns, to the financial detriment of the retailer (Shah et al. 2012). Therefore, it is important to look beyond retailers' return policies to consider the effectiveness of return prevention instruments.

Most past research reviews prevention instruments in isolation (see Table 1), which impedes understanding of the effectiveness of each instrument relative to other available tools. The majority of previous research has focused on monetary instruments (e.g., Anderson et al. 2009; Bower and Maxham 2012), with fewer studies examining customer-based (e.g., Bechwati and Siegal 2005) or procedural (e.g., Bonifield et al. 2010) instruments. In practice though, online retailers can choose from a variety of instruments and often apply several of them simultaneously (Bonifield et al. 2010; Douthit 2012; Lee 2015; Walsh et al. 2014). Moreover, the majority of past studies rely on cross-sectional survey data that may suffer from common method variance (Podsakoff et al. 2012) or else laboratory data that may lack realism (Wrenn et al. 2001); only a few offer field data (e.g., Zhou and Hinz 2016). Therefore, prior research cannot offer actionable insights for Internet managers.

To address these issues and extend previous work, we investigate the relationships of three return prevention instruments with return rates, using objective data from a large European online retailer. We purposely focus on one online retailer which allows us to compare the relative merit of each instrument. Moreover, objective field data offer benefits over cross-sectional or laboratory data, in that they capture actual shopping and return behavior. By using objective data to define the dependent and independent variables, we also can quantify the relative effectiveness of each instrument. The three instruments we investigate in this study are representative of three broad categories identified in prior literature: monetary, customer-based, and procedure instruments (e.g., Bonifield et al. 2010; Lee 2015; Walsh et al. 2014).

Specifically, we assess the effectiveness of three instruments: money-back guarantees, product reviews, and return labels/advice. These three instruments are representative of the categories that emerged from prior literature (see Table 2). In assessing them, we respond to Petersen et al.'s (2009, p. 102) calls for research into the "consequences of product return behavior." Drawing from risk theory, we propose that these three prevention instruments differentially affect an online retailer's return rate. To derive managerially relevant insights, we gather empirical results from three carefully controlled field experiments involving customers of an online retailer. These experiments show that the monetary instrument (money-back guarantee), unexpectedly, increases the retailer's return rate (experiment 1), whereas the customer-based instrument (product reviews) reduces it (experiment 2). The procedure instrument (return-related information, free return label) has no effect though (experiment 3). These findings are managerially relevant because they are actionable. Specifically, two of the instruments examined (i.e., money-back guarantee, return-related information and free return label) may be implemented with relatively little resources.

Background and hypotheses

Return-prevention instruments

Table 1 summarizes prior research into return-prevention instruments. According to Walsh et al. (2014), existing instruments consist of three main categories: monetary, customer-based, and procedural. Table 1 summarizes the result of our literature review which focused on articles published in leading journals identified from the Chartered Association of Business Schools (CABS 2015) ABS [including 4*, 3*] and from the German Academic Association for Business Research (VHB 2015) VHB3 [including A+,A, B]) journal lists. We searched for articles that contained keywords related to our research topic (e.g., "product returns", "preventive product returns", "product return prevention"). Table 2 gives examples of each instrument category.

The studies of the impact of monetary instruments tend to focus on restocking fees or money-back guarantees (e.g., Heiman et al. 2001; Hess et al. 1996; Suwelack et al. 2011). Essentially, monetary instruments provide a financial incentive to keep the product (Shulman et al. 2010; Walsh et al. 2014). For example, restocking fees can entice a consumer to keep the product rather than pay a financial penalty to return it (Shulman et al. 2011). Other measures may include giving customers a discount for not returning a product (Asdecker 2015).

Customer-based instruments are designed to inform customers about the product and processes before and during the purchase phase (Minnema et al. 2016; Walsh et al. 2014). More information, such as that provided through product reviews, avatars, or virtual try-on functions, might lead to better purchase decisions, based on more accurate product evaluations. For example, Zhou and Hinz (2016, p. 111) propose that "a better product description should reduce" the return rate.

Finally, procedural instruments aim to prevent product returns by improving the ordering and shipping processes, as well as making the return process as uncomplicated as possible, because these instruments also focus on post-purchase phases. The application of hassle costs represents an example (Mollenkopf et al. 2007); in general, hassle refers to the non-monetary effort associated with returning a product (Lambrecht and Tucker 2012; Mollenkopf et al. 2007). Making the process of returning a product inconvenient and tedious for customers may reduce their willingness to return products.

Table 1 Literature synthesis

Author	Description	Method/Sample	Theory	Studied Instruments
Anderson et al. (2009)	Model of customers' decision to purchase and return products	Field data, n = 987 customers	-	Monetary
Bechwati and Siegal (2005)	Framework of customer choice reversibility and likelihood of customer intentions to return products.	Laboratory experiments 1. Experiment: n = 87 2. Experiment n = 118	Inoculation theory	Customer-based
Bonifield et al. (2010)	Investigation of the relationship between return policy leniency and retailer quality.	Content analysis of online ratings, Rating site 1: n = 157 sites, site 2: n = 184 sites Experiment, n = 290	Signaling theory	Monetary, procedure
Bower and Maxham (2012)	Long-term consequences of fee and free return policies, as well as normative assumptions.	Longitudinal field study 1, n = 334; n = 351; n = 334 Longitudinal field study 2, n = 682; n = 1623; n = 1296	Attribution, regret, equity theory	Monetary
Heiman et al. (2001)	Conceptual, effect of MBGs and demonstrations for risk reduction.	-	-	Monetary, customer-based
Hess et al. (1996)	Investigation of the use of different charges to control product returns.	Observation, n = 28 online shops	-	Monetary
Janakiraman and Ordóñez (2012)	Investigation of the effects of time and effort on customers' decision to return a product.	Laboratory experiments, n = 276 students, n = 128 students, n = 153 students	Construal-level theory	Monetary, procedure
Lee (2015)	Investigation of product return behavior from a post-purchase perspective.	Survey, n = 61 consumers Survey, n = 83 consumers	Dissonance theory	-
McWilliams (2012)	Conceptual, optimal use of MBGs at high and low quality retailers.	-	-	Monetary
Minnema et al. (2016)	Investigation of the effect of product reviews on product returns for an electronic and furniture online shop.	Field data, n = 631,063 electronic and furniture purchases	-	Customer-based
Ofek et al. (2011)	Development of a theoretical model to analyze how a manufacturer coordinates channels in pricing and return policies.	-	-	Monetary
Petersen and Kumar (2009)	Investigation of effects of product returns on customer buying behavior and firm-customer exchange process.	Industrial, historical sample, n = 1179, n = 393	-	Monetary
Petersen and Kumar (2015)	Investigation of the interaction between perceived risk, product returns and optimal resource allocation.	Industrial, historical sample, n = 26,000	Risk theory, commitment–trust theory	Monetary
Shah et al. (2012)	Investigation of the profitability of cross-buying behavior with consideration of return policy leniency in one observed firm.	Field data, n = 147,901 retail customers, n = 360,819 business, government and educational institution customers, n = 1025 SME and large business organization customers, n = 92,155 customers, n = 32,389 customers	-	-
Shulman et al. (2011)	Theoretical, optimal pricing and restocking fee decisions of two competing retailers.	-	-	Monetary Monetary

Table 1 (continued)

Author	Description	Method/Sample	Theory	Studied Instruments
Suwelack et al. (2011)	Investigation of cognitive, affective, and behavioral outcomes of the use of MBGs.	Laboratory experiments, $n = 200$, $n = 400$	Risk theory, equity theory	
Zhou and Hinz (2016)	Modeling of the purchase and return behavior, focusing on return policies and reputation.	Field data, $n = 592$ shops	-	Monetary, Customer-based

To our knowledge, Walsh et al. (2014) assertion that the three categories of instruments are appropriate preventers of product returns has never been empirically tested.

Hypotheses development

Consumers experience perceived risk when they shop online (Forsythe and Shi 2003; Mollenkopf et al. 2007; Yalabik et al. 2005), because they cannot evaluate the products in advance and often must pay for the purchased products before shipment. Perceived risk refers to the consumer's personal assessment of the risk associated with a decision or action, typically a purchase decision (Mitchell 1999). Risk theory posits that the consumer's perceived risk can be financial, social, or physical or some combination (Laroche et al. 2004). We believe risk theory is suitable to the study of online shopping behavior. Because it is not possible to physically examine and touch products before making an online purchase (Shulman et al. 2010), consumers likely perceive risk when shopping online and assessing retailers' return prevention instruments (Forsythe and Shi 2003; Heiman et al. 2001). Typically, prior literature distinguishes five risk dimensions: financial, time, performance, social, and psychological (e.g., Laroche et al. 2004). For this study, perceived risk refers to a subjective expectation of loss (Laroche et al. 2004; Mitchell and Greatorex 1993). We posit that each of the three return-prevention instruments is associated with varying levels of risk, such that they vary in their effectiveness for preventing returns.With a money-back guarantee (MBG), consumers have the option of returning the product during a defined time period (e.g., four weeks) and receiving a refund (Heiman et al. 2001). In the EU, an MBG (also referred to as 'satisfaction guarantee') usually begins after the legal cancellation period of 14 days. Within that period customers have the right to cancel and return their order and to be refunded. The MBG allows the customer to return a product, and get a refund, beyond the legally binding period, for example, within 30 days.

Because MBGs offer a signal of quality (Moorthy and Srinivasan 1995), Heiman et al. (2001) maintain that they are effective for reducing consumers' perceived risk prior to ordering. Similarly, Zhou and Hinz (2016, p. 104) argue that MBGs are part of a broader lenient return policy and that lenient "return policies act as a signal that induces customers to perceive higher quality and lower risk".

Furthermore, MBGs can protect consumers against unsatisfactory product performance and personal misfit by removing the financial risk for the customer (Heiman et al. 2001). They accordingly increase consumers' willingness to buy new products (Davis et al. 1995) and premium products (Suwelack et al. 2011). Risk theory suggests that MBGs even allow consumers to buy products without any financial risks. Similar to retailer reputation, a MBG can signal implicit quality, which reduces customer perceived risk and affects customer behavior beyond the purchasing act (Walsh et al. 2016). For example, if a customer buys a fashionable new handbag online, she might be uncertain about whether the handbag will look good with her existing apparel and the resulting financial risk, such

Table 2 Product return prevention instruments (based on Walsh et al. 2014)

Monetary instruments	Customer-based instruments	Procedural instruments
- Restocking fees	- Virtual try-on	- Safety packaging
- Money-back guarantee	- Avatar	- Cycle-time optimization
- Discounts (for not returning products)	- Customer reviews	- Return advice
- Gifts (for not returning products)	- Product advice	- Different return channels
- Prepayment methods	- Height/size charts	- Contacting/banning of "return sinners"
- Restriction of the order volume	- - Product-availability information	- Not providing merchandise return card
- Shipping costs	- Customer hotline	- Final package inspection
	- Alternative product photos	- Personalized package extras (e.g., leaflets)
	- Zoom technologies	

as payment in advance via credit card. Therefore, she is more likely to order the handbag if the online shop provides a MBG, which decreases the chances of product returns, because the customer's financial risk is lessened. In turn, we assume a positive relation between the implementation of an MBG and the product return rate.

Hypothesis 1: The product return rate is lower in the presence, rather than the absence, of an MBG.

Customer-based instruments aim to provide the customer with more information to evaluate the product's suitability and fit (Walsh et al. 2014). Reviews are a virtual source of customer opinions about and experiences with products (Weathers et al. 2015), so other customers use this instrument to improve their decision making about the product (Mudambi and Schuff 2010; Sahoo et al. 2015; Weathers et al. 2015). Product reviews are available from non-retail websites, like Yelp.com or Ciao.com, as well as from online retail sites (e.g., Amazon.com) (Weathers et al. 2015). As prior research shows, they are effective in influencing sales and product choices (Chevalier and Mayzlin 2006; Senecal and Nantel 2004). Minnema et al. (2016) report that an overly positive valence in product reviews might lead to more returns though, because above-average and favorable reviews induce purchases but also increase return likelihood. However, their study does not account for differences between the implementation versus no implementation of product reviews.

Consistent with prior research (e.g., Hennig-Thurau et al. 2004), we argue that product reviews help consumers in their decision making. Perceived risk is likely especially high when buying online and particularly acute in categories where consumers need to touch and feel the product in order to determine how well it fits their tastes and needs. Accordingly, Zhu and Zhang (2010, p. 135) maintain that "online reviews could be useful in reducing the risk of purchasing such products". If product reviews are available to consumers when they make purchase decisions, they should not only reduce perceived risk but also contribute to more informed, better buying decisions. Sound buying decisions are less likely than poor decisions to prompt post-purchase dissonance, which can instigate product returns (Walsh et al. 2016).

Hypothesis 2: The product return rate is lower in the presence, rather than the absence, of product reviews.

Consumer perceived risk refers to subjective perceptions of the possible negative consequences that a behavior, such as a purchase, might produce (Büttner and Göritz 2008). We theorize that returns-related information provided to (prospective) customers should reduce their pre-purchase risk and induce product returns. Many online retailers such as The Gap, Nike, Asos, Zalando, and Otto also rely on efficient product return procedures, such as providing a free return shipping label and a short description of the product return process included with every order, to make the product return procedure as hassle-free as possible (Bonifield et al. 2010). The label and description likely facilitate the decision to return in situations in which the consumer is not entirely sure if the ordered product meets expectations. Because the label and description represent a signal of trustworthiness to the customer, they induce the customer to perceive less risk which likely results in a higher shopping volume and return rate. Without such aids, the return process remains complex, so the consumer must incur search costs to find, order, or print out a return label, as well as to contact the retailer to find out how the product can be returned. These hassle costs and the concurrent time risk may cause consumers to keep the ordered product, even it does not fully meet their expectations.

Hypothesis 3: The product return rate is lower in the absence, rather than the presence, of an explanation of the product return process and free shipping label.

Method and Results

Procedure and data

To test the hypotheses, we conducted three field experiments in the online shop of a well-known premium apparel manufacturer and retailer that is headquartered in Germany and earns turnover in the hundreds of millions Euros. The return rate from customers is less than that for multisegment apparel retailers like Zalando or Otto but still consistently around 20%. The experiments were run independently, to minimize any potential spillover effects or biases, between 2013 and 2015.

To observe return behavior, we chose return rates as a quantitative measure. We calculated the product return rate according to the number of ordered and returned articles for each shipment, as is very common in practice (Young 1988). Specifically,

$$\text{return rate} = \frac{\text{returned articles } \textit{per} \text{ order}^{*}}{\text{ordered articles } \textit{per} \text{ order}} 100\% \qquad (1)$$

If a customer keeps all ordered articles, the return rate is 0%; if all ordered articles are returned, the return rate is 100%.

The first field experiment investigated the impact of a MBG on customers' return behavior. The MBG was displayed on every shop page in the banner, as well as in a special advertisement in the electronic shopping basket (see Appendix 1). We implemented the experiment in two phases: First, we re-configured the web shop to contain a MBG in different places on the website with a "Big Bang" strategy (O'Leary 2000), so all the manipulations were active in the real web shop. Second, after two weeks, we returned the web

shop to a non-MBG and measured the difference. Furthermore, we confirmed the conditions were similar to those prior to the implementation of the MBG by checking average return rates and the same range of products. An overview of the experiments is in Table 3.

To explore the effect of product reviews on product return behavior, we discussed the experimental design with the web shop management prior to initiating the experiment. A specifically designed review system was implemented that allowed customers to evaluate the product using a short review mechanism and a 5-star rating system (0 = very bad product, 5 = very good product). This system is comparable to reviews in other online marketplaces (Mudambi and Schuff 2010), as illustrated anonymously in Appendix 2. Before the experimental implementation, the online shop provided no customer reviews. The experimental group was defined by all purchases within a specific time period in which at least one product appeared with reviews. The customers in the control group purchased during the same time period, but the products they purchased were presented in the online shop without any product reviews.

Finally, for the test of Hypothesis 3, we noted that prior to the experiment, every package shipped by the retailer contained both the ordered articles and a separate sheet detailing the return process, as well as a free label for returning the products. During the experimental phase, the retailer abandoned the provision of the instruction sheet and shipping labels, such that customers had to obtain the information and labels online, via the the web shop or email, or request them via telephone.

In each experiment, the data analysis involved a comparison of the data obtained from the experimental and control groups. Because the experiments provided objective data, we had to expect data with a skewed (asymmetrical) distribution. When the assumption of normality is questionable, a simple t-test is inappropriate. Instead, the Mann-Whitney U-test should be employed, which is a non-parametric test for two independent groups (Grover and Vriens 2006). The Mann-Whitney U-test is commonly used in scholarly research (Palacios Fenech and Tellis 2016; Soh et al. 2006).

Table 3 Overview of field experiments

	Experimental Group	Control Group
Experiment 1 (monetary instrument: MBG)		
Number of ordered articles	1143	1036
Returned articles	127	90
Return rate	11.11%	8.68%
p	0.047	
Experiment 2 (customer-based instrument: product reviews)		
Number of ordered products:	660	1376
Returned products:	25	78
Return rate	3.78%	5.66%
p	0.042	
Experiment 3 (procedure instrument: return information and free return labels)		
Number of ordered products	1106	1124
Returned products	89	82
Return rate	8.04%	7.29%
p	0.825	

Results

Experiment 1 The first field experiment investigated the influence of the use of a MBG on customers' product return behavior. According to the Kolmogorov-Smirnov-test ($p < 0.01$) and Shapiro-Wilks test ($p < 0.01$), the data were not normally distributed, so we used a Mann-Whitney U-test to test the hypothesis. This test revealed that the product return behavior of the two groups differed significantly ($p = 0.047$), but the mean value of the return rate increased from 8% to 11.11% in the presence of a MBG, in contrast with Hypothesis 1. The time period between buying and returning a product also increased, from 6.3 days to 7.8 days. These unexpected results reveal that MBGs drive product returns. Perhaps MBG signal not only product quality to encourage the purchasing act (which may induce customers to retain a product even if they are not entirely sure it meets their expectations) but also retailer quality, which encourages customers to order more (due to the low perceived risk afforded by the MBG). Ultimately, these two forces might lead to higher, not lower, return rates.

Experiment 2 This field experiment examined the effect of product reviews on product return behavior, and again, the data were not normally distributed, according the Kolmogorov-Smirnov test ($p < 0.01$) and Shapiro-Wilks test ($p < 0.01$). The Mann-Whitney U-test test shows that the two groups differ significantly in their product return behavior ($p = 0.042$): The mean percentage return rate in the absence of product reviews (5.66%) is higher than that in the presence of product reviews (3.78%), consistent with Hypothesis 2. That is, product reviews can decrease product returns.

Experiment 3 The influence of a more difficult return process, involving the omission of return-relevant information and no free return label, was analyzed in the third field experiment. These data were not normally distributed either, according to the Kolmogorov-Smirnov test ($p < 0.01$) and the Shapiro-Wilks test ($p < 0.01$). The Mann-Whitney U-test showed no significant differences between the product return rates of the experimental and control groups though ($p = 0.825$). A consumer-friendly return process has no influence on product return behavior, and we must reject Hypothesis 3.

Discussion

For online retailers, it is important to implement preventive product return management instruments to reduce costs and ensure a strategy-aligned customer orientation. When customers shop online they must deal with the lack of potentially pertinent information and tactile cues (Shulman et al. 2010). Therefore, different types of risk arise for online purchasing and returns (Heiman et al. 2001; Mitchell 1999). The current research builds on risk theory to demonstrate how different return-prevention instruments increase or dampen product return behavior. Previous empirical research has mainly considered instruments in isolation (McWilliams 2012; Minnema et al. 2016); some studies also theorize about the effectiveness of different instruments (Bonifield et al. 2010; Lee 2015; Walsh et al. 2014) without assessing them according to empirical or experimental field data. Our research shows that three common return-prevention instruments have varying impacts on customer product return rates. With experimental field studies, we investigate the effectiveness of instruments representative of monetary, customer-based, and procedural return-prevention instruments: money-back guarantees, product reviews, and missing return information or free return labels within the package. The field experiments show that product reviews can reduce product return rates, whereas MBGs have a return-increasing effect. Furthermore, explanations of the return process and enclosing a free return label does not affect customers' return behavior. Nonetheless, this is an important finding because omitting this instrument could save the retailer the costs associated with providing such information and labels.

Implications

Our field experiments show how different return-prevention instruments influence customer product return rates and thereby contribute to nascent literature on preventing returns for online retailing (e.g., Bonifield et al. 2010; Hess et al. 1996; Powers and Jack 2015; Shulman et al. 2010) by adding new insights about the effectiveness of three product return prevention instruments, in terms of reducing customer product return behavior. First, MBGs have negative effects; they actually increase the rate of product returns by 41%. This is a noteworthy finding, not least because product returns are a cost factor for the shop we partnered with for this research. Every product return results in costs of 10–15% of the ordered products on average. Therefore, applying a MBG increases the handling costs of the online shop. From a consumer perspective, MBGs enhance online shopping convenience. But for online retailers, they are unfavorable because they drive product returns. Our data suggests that for the shop used in the experiments the use of this instrument negatively affects profitability. This is an unexpected yet interesting finding. Economic theory would suggest that in the absence of tactile information consumers make a greater decision-making effort (Citrin et al. 2003; Peck and Childers 2003), presumably resulting in 'better' decisions and less returns. However, MBGs seem to entice the customer to make less thorough decisions, thus explaining the ensuing higher return rate.

Therefore, an MBG should be used only for products associated with high uncertainty that otherwise would not be purchased online. In particular, online retailers should assess whether the higher (return-related) costs associated with MBGs are offset by higher order volumes and revenues. Recent research by Zhou and Hinz (2016) suggests that lenient return policies can lead to higher returns and improved profitability. Thus, it would interesting to explore under what conditions MBGs would have a profitability-enhancing effect.

Second, with customer-based instruments like product reviews, customers can better evaluate the product, which tends to minimize their product return rates. This decreased product return behavior is favorable from a retailer perspective. The second field experiment showed that the product return rate decreases by 49%. The saved costs increases profitability of the online shop increased. We thus extend previous research (e.g., Minnema et al. 2016) by investigating return behavior in shopping situations in which product reviews are implemented versus not implemented. Our findings are broadly consistent with prior research that shows that product reviews have risk-reducing and spending-increasing effects in online shopping settings (e.g., Bae and Lee 2011; Wu et al. 2013). Online retailers should therefore consider embedding online reviews on their website. Before doing so, however, they should weigh the costs of implementing and maintaining an online review system against the costs of product returns. Furthermore, they should take care about the challenges in relation with fake reviews and unauthorized content. Third, packages without free return labels and no explanation of the return process could incur high hassle costs for customers, because the return procedure becomes less transparent and more tedious. But our results show that this instrument still does not affect product return behavior. Instead, it implies some savings potentials for the retailer, in terms of the cost and time required to insert the instructions and provide free shipping labels. To be more precise, the retailer can save costs and time in terms of printing out the label and return description. Additionally, there are time and cost savings in relation to the (manual) process of adding the documents to the package.

With these findings, we build on and extend research by Walsh et al. (2014), who detail various return-prevention

instruments but do not assess them empirically. We implement the different instruments in a real-world setting, with the cooperation of a well-known European online store, to test their effectiveness. Notably, the online store did not receive any customer complaints or identify any measurable impacts on buying behavior associated with implementing these experiments, even the one involving the omission of return information and a free preprinted return label. During the period of the experiment, the online store saved money (i.e., ten euro-cents per shipped package) by eliminating the labels and return information. Therefore, the online store is reconsidering its use of this instrument, because eliminating it might save costs without any detriment in terms of return behavior.

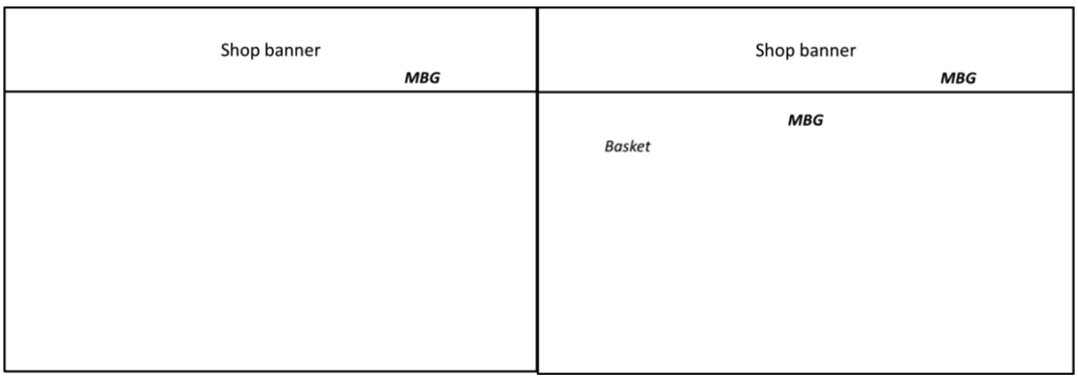
Limitations and further research

Several study limitations and resulting areas for further research deserve mention. Our three field experiments took place on the website of one, well-known European apparel retailer. Although we respond to the call for more research using field experiments (Gneezy 2017), we acknowledge that our results may suffer from limited external validity. Furthermore, the retailer that agreed to participate in this study may be different from those that did not (Heckman 1992). Therefore, the results might not generalize to online retailers in other product categories. Furthermore, the three instrument groups may differ in effectiveness across countries; non-EU retailers may work under different return-related regulations that could affect the viability and effectiveness of these instruments. Further research should continue to test the effectiveness of different return-prevention instruments across various online retailers and countries. Moreover, we only studied transactions (i.e., buying and returning) in limited shopping periods, lasting a few weeks, largely due to the time and costs associated with implementing the field experiments. It would be interesting to assess the effectiveness of the return-prevention instruments relative to multiple transactions or over time as well as the effect of those instruments on consumer behavioral outcomes (Walsh and Brylla 2016). We used the actual return rate to measure customer product return behavior (Young 1988); return-related word of mouth behavior also could be of interest, as a means for retailers to predict customers' future purchases. In terms of the monetary instrument of MBGs, we implemented a reward instrument, which tends to increase (not decrease) the return rate. Additional research could examine the effectiveness of monetary instruments that adopt a punitive character, such as restocking fees (Hess et al. 1996).

Finally, we did not discuss profitability in relation to the three tested instruments and product returns in greater depth because comprehensive profitability data was not made available to us. Such data may have allowed us to reconcile some conflicting findings in previous studies. For example, Petersen and Kumar (2010) report that a lenient return policy can cause reductions to profit of almost 4%. In contrast, Zhou and Hinz (2016) show that retailers that make their return policy more lenient can increase their profits by 0.016%. Future research should aim to clarify to the relationship between return policies, return prevention instruments, and firm profitability.

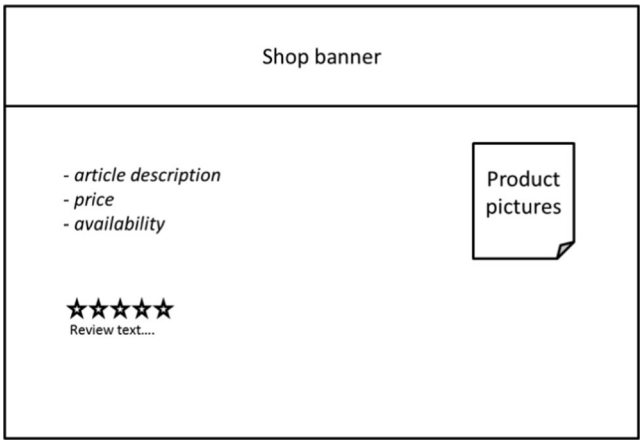
Appendix 1

Abstract illustration of the placement of the MBG in the experimental shop



Appendix 2

Abstract illustration of the product reviews



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